

THIRUVALLUVAR UNIVERSITY
SRI BHARATHI WOMEN'S
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Topic:

Farm Management System

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FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

2.2.2 Empathy Map Canvas

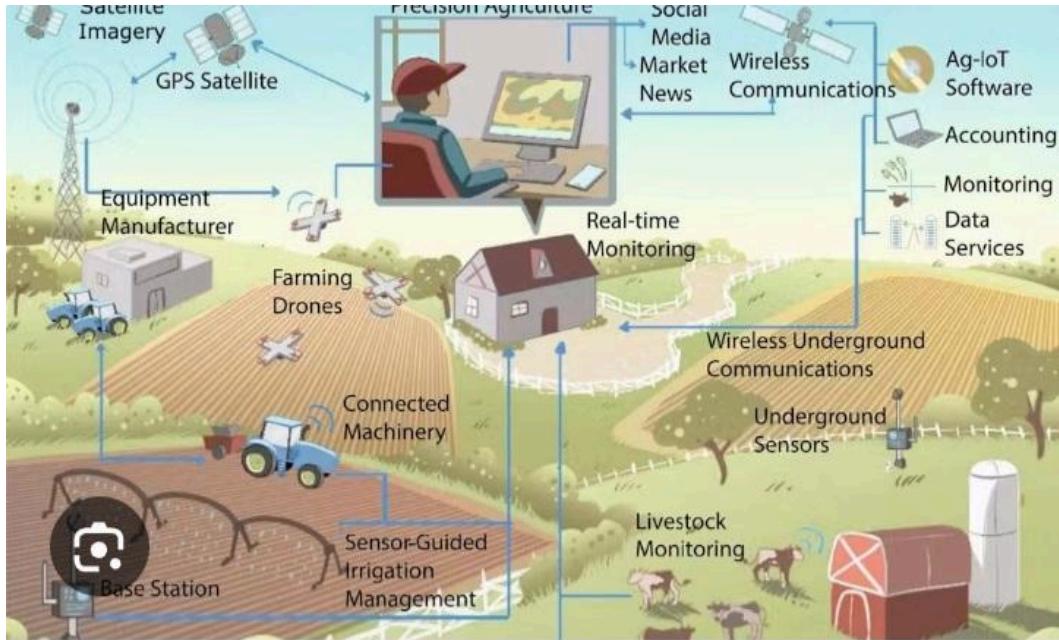
Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets

- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



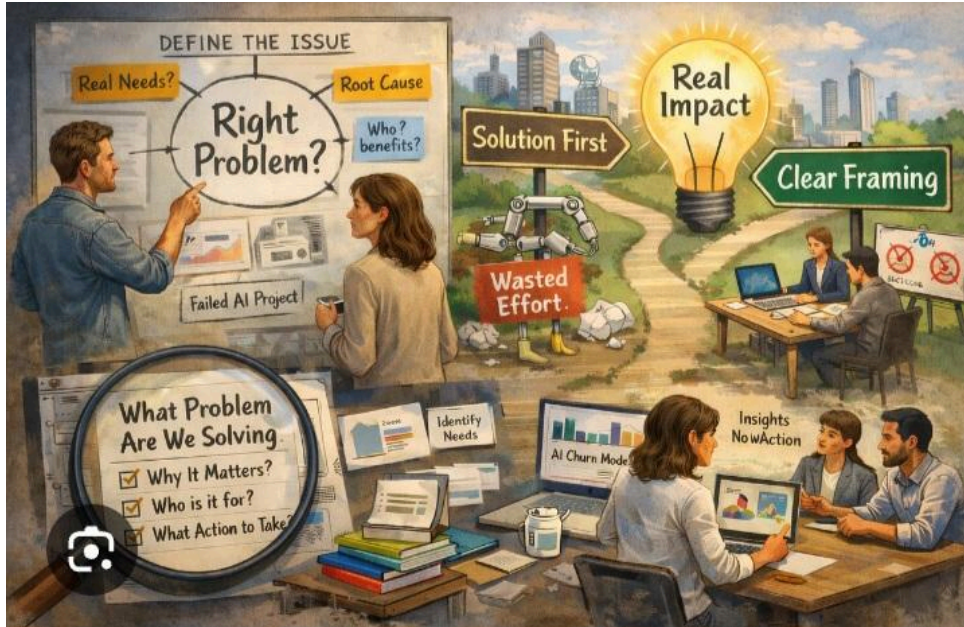
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

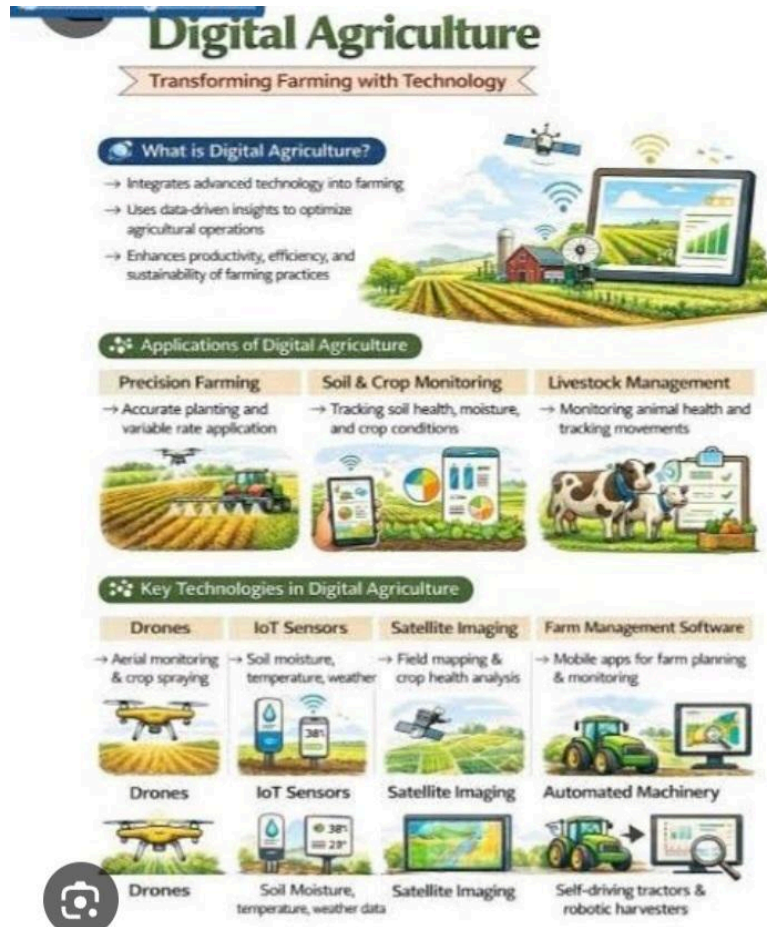


3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping

FR-5	Report Generation
FR-6	Dashboard Visualization
FR-7	User Role Management

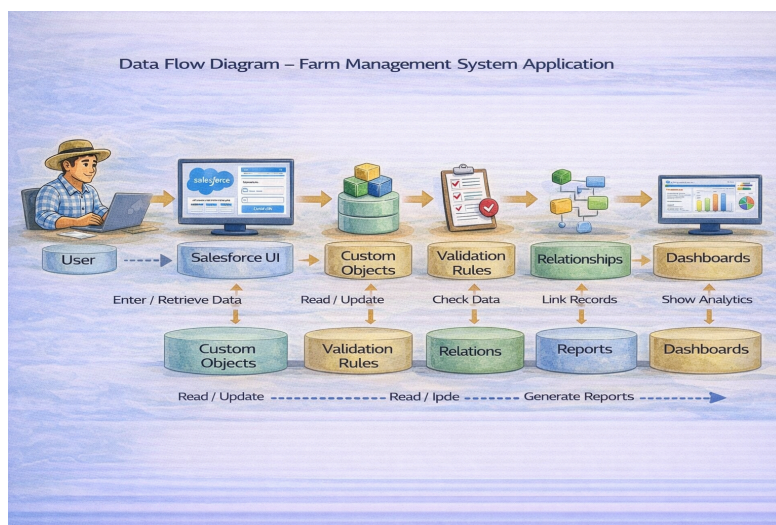
Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



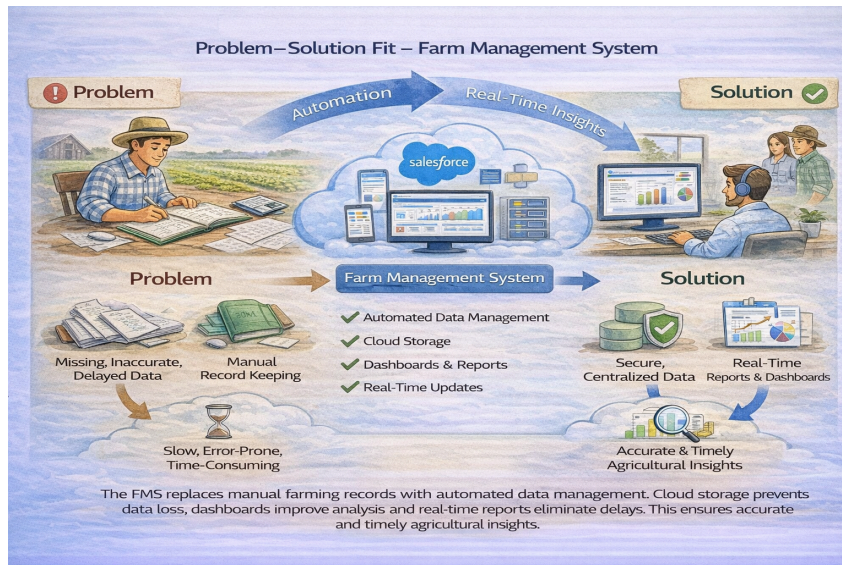
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



4.2 Proposed Solution

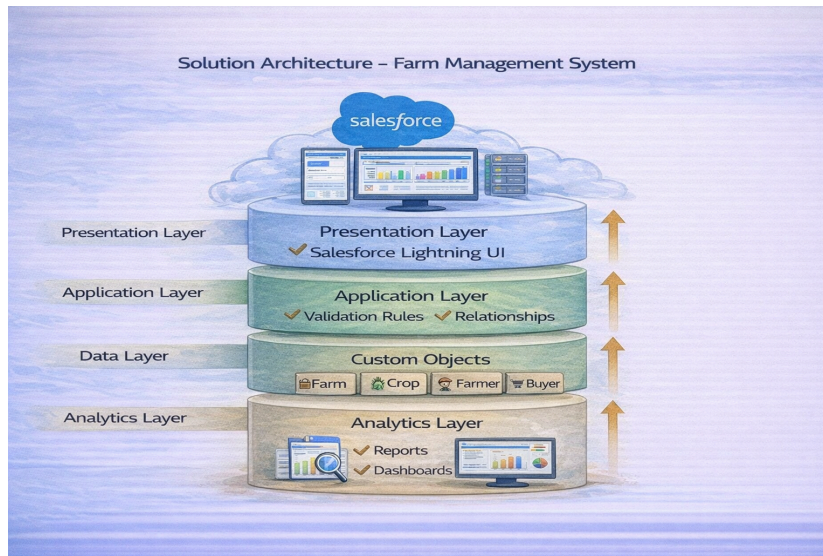
Functional Requirements

Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.
Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer	5	High	Member2

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			objects with proper relationships so that agricultural data is stored and managed efficiently.			
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	2 Days	19 Sep 2026	20 Sept 2026	08	17 Sep 2026
Sprint-2	20	2 Days	21 Sep 2026	22 Sep 2026		
Sprint-3	20	2 Days	23 Sep 2026	24 Sep 2026		
Sprint-4	20	2 Days	25 Sep 2026	26 Sep 2026		

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will receive a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



Build enterprise-quality apps fast to bring your ideas to life

Build apps fast with drag-and-drop tools
Customize your data model with clicks
Go further with Apex code
Integrate with anything using powerful APIs
Stay protected with enterprise-grade security
Customize UI with clicks or any leading-edge web framework



Sign up for your Salesforce Developer Edition A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*	Last Name*
Email*	
Role*	
Company*	
Country/Region*	
Postal Code*	
Username*	

Your username must be in the form of an email address (it does not have to be real). It must be unique and cannot be associated with another Salesforce login credential. [Read more about username recommendations.](#)

☐ I agree to the [Main Services Agreement](#) – Developer Services and [Salesforce Program Agreement](#).

Sign me Up

6. By using username and password you can into the Salesforce org. The

homepage will appear below

Setup

Home

Object Manager

Search Setup

Quick Find

Setup Home

Service Setup Assistant

Commerce Setup Assistant

Hyperforce Assistant

Release Updates

Lightning Experience Transition Assistant

Salesforce Mobile App

Lightning Usage

Optimizer

Sales Cloud Everywhere

ADMINISTRATION

> Users

> Data

> Email

PLATFORM TOOLS

> Subscription Management

> Apps

> Feature Settings

> Slack

> Heroku

> MuleSoft

SETUP

Home

Create



Get Started with Einstein Bots

Launch an AI-powered bot to automate your digital connections.

Get Started



Mobile Publisher

Use the Mobile Publisher to create your own branded mobile app.

Learn More



Real-time Collaborative Docs

Transform productivity with collaborative docs, spreadsheets, and slides inside Salesforce.

Get Started

Most Recently Used

10 Items

NAME	TYPE	OBJECT
SLA Serial Number	Custom Field Definition	Account
Upwell Opportunity	Custom Field Definition	Account
Active	Custom Field Definition	Account
SLA Expiration Date	Custom Field Definition	Account

Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

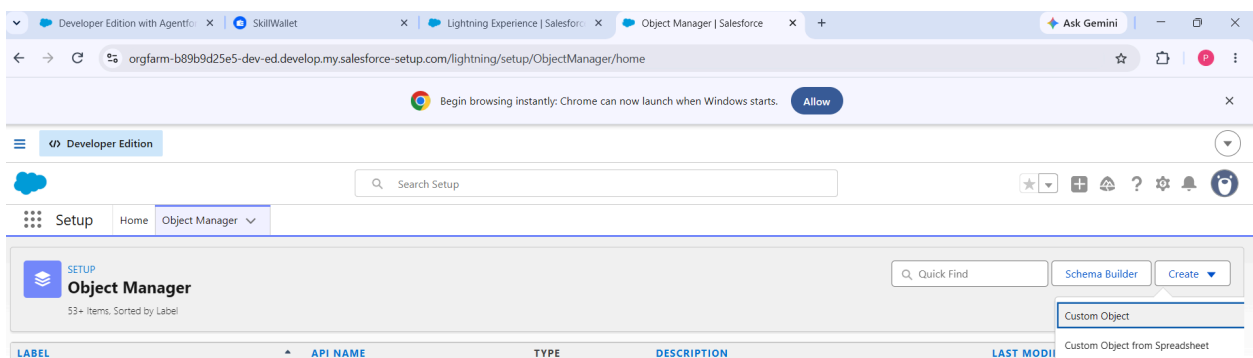
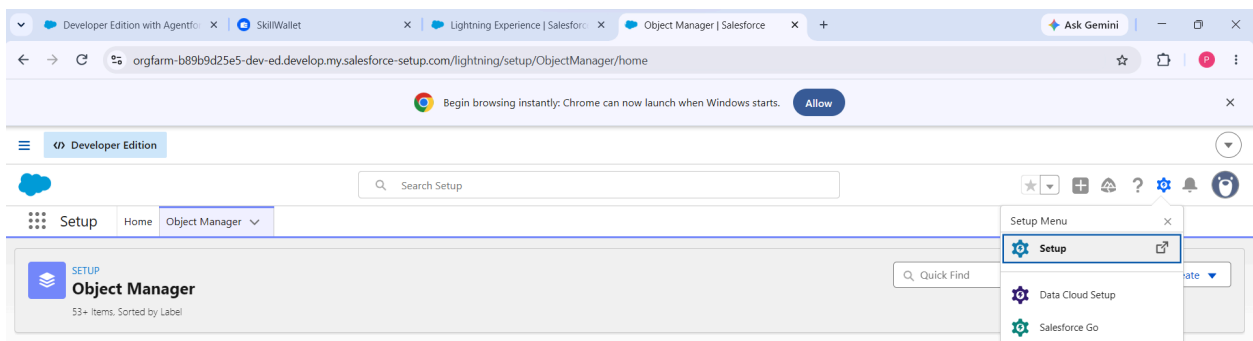
- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"

6. Select the data type as "Text".

The screenshot shows the Salesforce Object Settings page for a new object. The left sidebar contains a navigation menu with the following items: Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled "Data Type" and shows a dropdown menu set to "Text". A warning message states: "Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type." Below this, there are three sections: "Optional Features", "Object Classification", and "Deployment Status". The "Optional Features" section has checkboxes for "Allow Reports" (checked), "Allow Activities" (unchecked), "Track Field History" (unchecked), "Allow in Chatter Groups" (unchecked), and "Enable Licensing" (unchecked). The "Object Classification" section has a note: "When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. Learn more." It has checkboxes for "Allow Streaming" (checked), "Allow Bulk API Access" (checked), and "Allow Streaming API Access" (checked). The "Deployment Status" section has radio buttons for "In Development" (unchecked) and "Deployed" (checked). Below this is the "Search Status" section with a note: "When this setting is enabled, your users can find records of this object type when they search. Learn more." It has a checkbox for "Allow Search" (checked). At the bottom right, there are three buttons: "Save", "Save & New", and "Cancel".

7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list

to default page layout.

11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer
 - c. Plural Label: Farmers
 - d. Record Name: Farmer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list

to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Milestone 3 - Tab: Tab

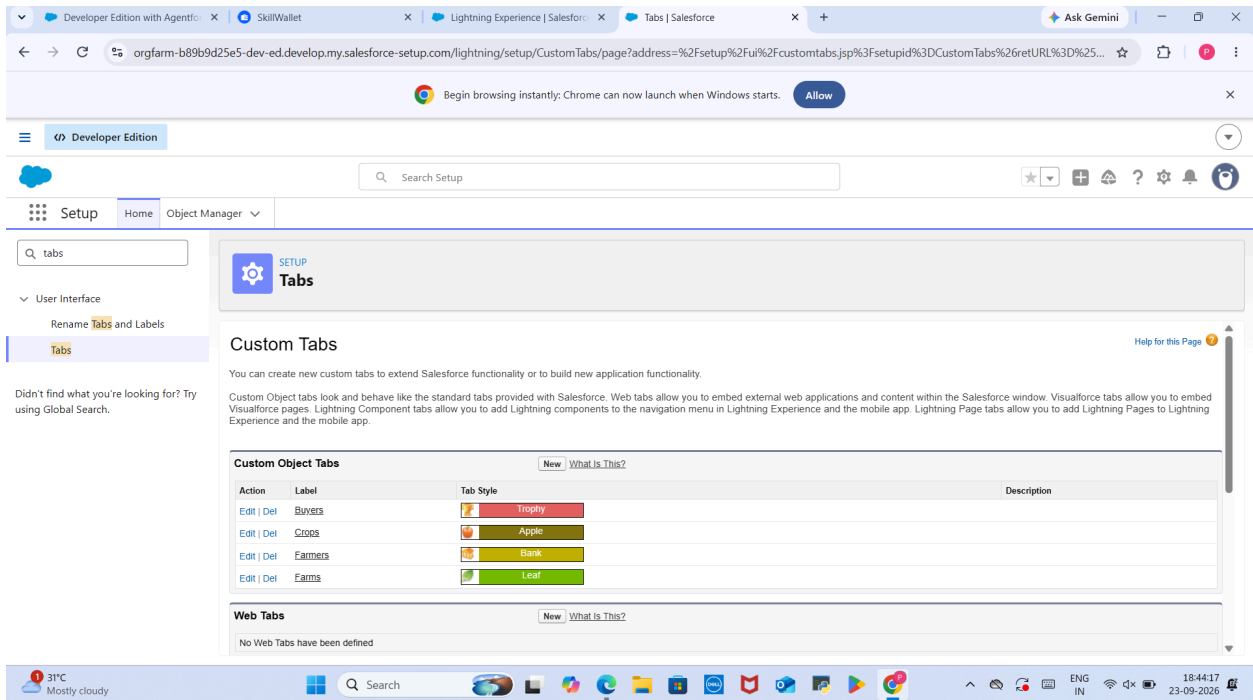
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

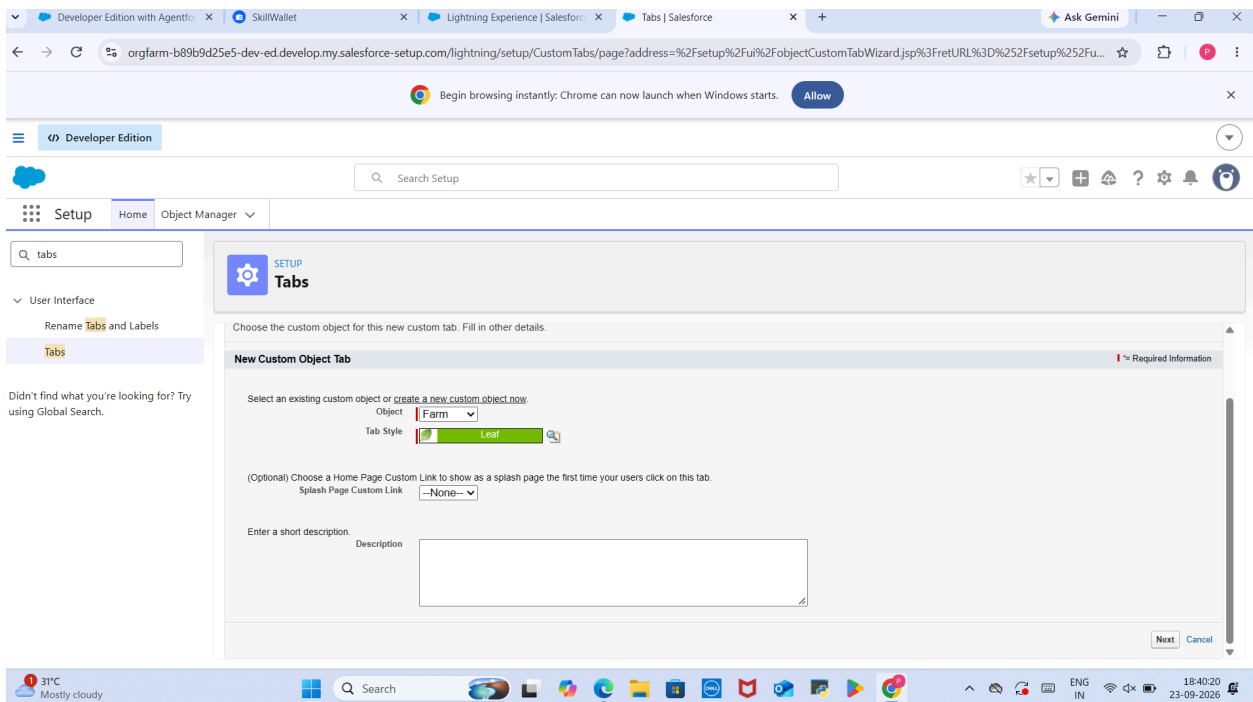
Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. Under custom object tabs, click New.



3. For Object, select Farm.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

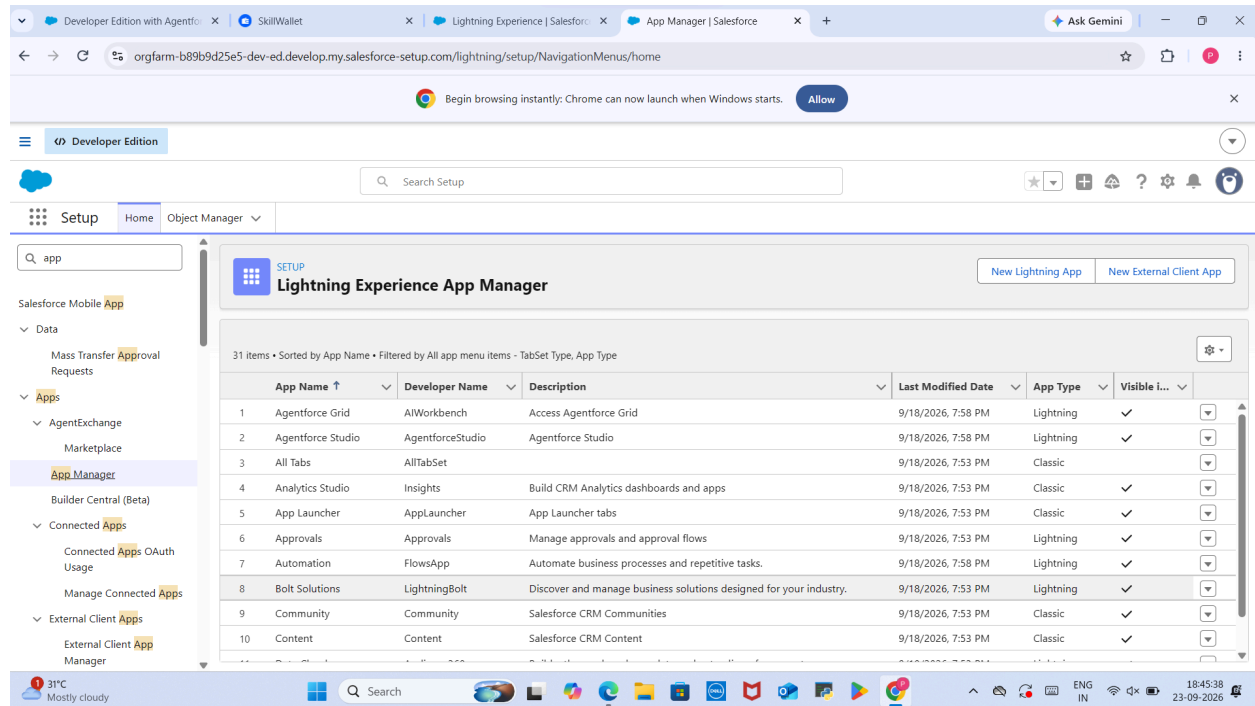
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app



- From Setup, enter App Manager in the Quick Find and select App Manager.
- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next.
- From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.
- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

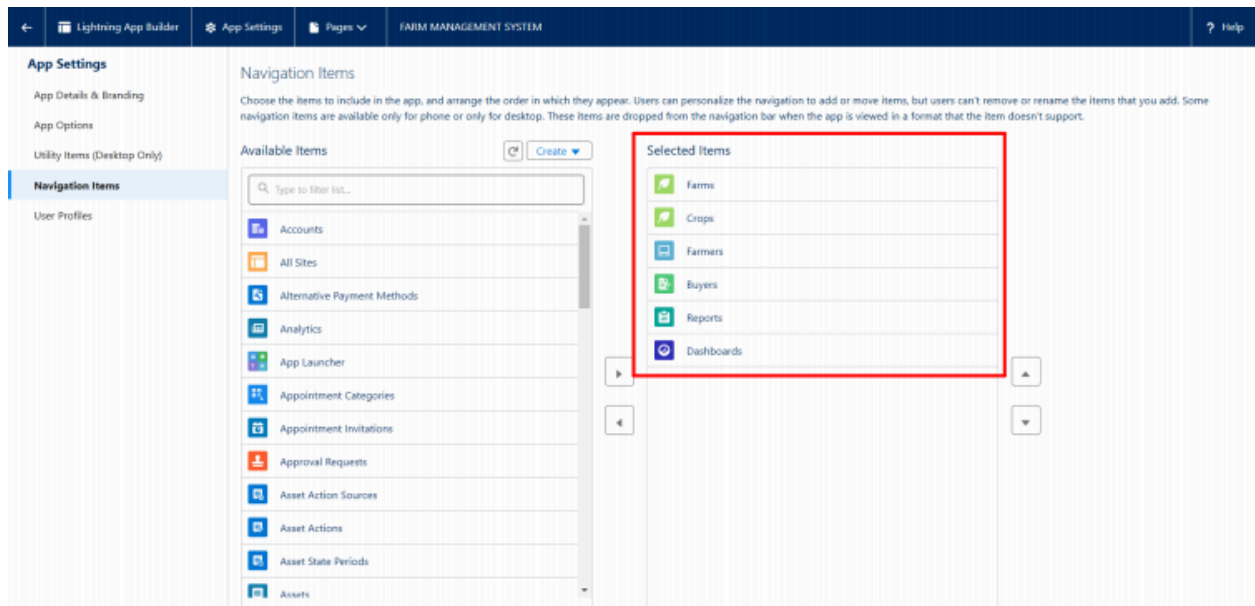
1. App Launcher-Displays available apps.

2. App Name-Displays the current selected app.

The screenshot shows the 'New Lightning App' setup window in Salesforce. The main heading is 'App Details & Branding'. Below this, there are two columns: 'App Details' and 'App Branding'. In the 'App Details' column, the 'App Name' field is highlighted with a red box. Below it are fields for 'Developer Name' and 'Description'. In the 'App Branding' column, there is an 'Image' upload section and a 'Primary Color Hex' field with a dropdown menu. At the bottom right of the window, the 'Next' button is highlighted with a red box. Below the setup window, a table lists existing Lightning apps.

ID	App Name	Category	Description	Created Date	Theme	Status
13	Lightning Usage App	LightningInstrumentation	View Adoption and Usage Metrics for Lightning Experience	11/04/2024, 12:12 pm	Lightning	✓
14	Marketing CRM Classic	Marketing	Track sales and marketing efforts with CRM objects.	11/04/2024, 12:12 pm	Classic	✓
15	Platform	Platform	The fundamental Lightning Platform	11/04/2024, 12:12 pm	Classic	✓

3. Navigation menu-Displays the tabs available inside the app.



Milestone 5 -

Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New

The screenshot shows the Salesforce Setup interface for the 'Farm' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, and Search Layouts. The main content area is titled 'Fields & Relationships' and shows a table of 4 items, sorted by Field Label. The table has columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The items listed are: Created By (CreatedById, Lookup(User)), Farm Name (Name, Text(80)), Last Modified By (LastModifiedById, Lookup(User)), and Owner (OwnerId, Lookup(User,Group)).

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Farm Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓

5. Choose Data type as a picklist, Click next

The screenshot shows the Salesforce Setup interface for a 'Farm' object. The left sidebar contains a navigation menu with the following items: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, and Search Layouts. The main content area is titled 'Farm' and displays a list of data types with their descriptions. The 'Picklist' option is selected, and its description is shown on the right.

Data Type	Description
☐ Email	Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.
☐ Geolocation	Allows users to define locations. Includes latitude and longitude components, and can be used to calculate distance.
☐ Number	Allows users to enter any number. Leading zeros are removed.
☐ Percent	Allows users to enter a percentage number, for example, '10' and automatically adds the percent sign to the number.
☐ Phone	Allows users to enter any phone number. Automatically formats it as a phone number.
☒ Picklist	Allows users to select a value from a list you define.
☐ Picklist (Multi-Select)	Allows users to select multiple values from a list you define.
☐ Text	Allows users to enter any combination of letters and numbers.
☐ Text Area	Allows users to enter up to 255 characters on separate lines.
☐ Text Area (Long)	Allows users to enter up to 131,072 characters on separate lines.
☐ Text Area (Rich)	Allows users to enter formatted text, add images and links. Up to 131,072 characters on separate lines.
☐ Text (Encrypted)	Allows users to enter any combination of letters and numbers and store them in encrypted form.
☐ Time	Allows users to enter a local time. For example, "2:40 PM", "14:40", "14:40:00", and "14:40:50.600" are all valid times for this field.
☐ URL	Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)

7. Next, Next Click Save.

The screenshot shows the Salesforce Setup interface for creating a new custom field for the 'Farm' object. The browser address bar shows the URL: `orgfarm-b89b9d25e5-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lgK000007Y1Y9/FieldsAndRelationships/new`. The page title is 'Farm' under 'SETUP > OBJECT MANAGER'. The left sidebar shows the navigation menu with 'Fields & Relationships' selected. The main content area is titled 'New Custom Field' and 'Step 2. Enter the details'. The 'Field Label' is 'Farm Type'. The 'Values' section has two options: 'Use global picklist value set' (selected) and 'Enter values, with each value separated by a new line'. The 'Field Name' is 'Farm_Type'. The 'Description' is empty. The 'Einstein' logo is visible in the bottom right corner of the page. The Windows taskbar at the bottom shows the date and time as 18:56:40 on 23-09-2026.

Developer Edition with Agenti x SkillWallet x Lightning Experience | Salesfo x Recently Viewed | Buyers | Sal x Farm | Salesforce x + Ask Gemini x

orgfarm-b89b9d25e5-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lgK000007Y1Y9/FieldsAndRelationships/new

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Farm

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

Farm

New Custom Field

Step 2. Enter the details

Step 2 of 4

Field Label Farm Type

Values

☒ Use global picklist value set

☐ Enter values, with each value separated by a new line

--None--

☐ Display values alphabetically, not in the order entered

☐ Use first value as default value

☒ Restrict picklist to the values defined in the value set

Field Name Farm_Type

Description

Einstein

31°C Mostly cloudy

Search

ENG IN

18:56:40 23-09-2026

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

- 1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
- 2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
- 3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
- 5.Create a Master Detail relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

- 1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
- 3.Again we select data type as a date,Enter field labe (Harvest date),save
- 4.Next We Create Lookup relationship on crop object realted to Farm Object.
- 5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.
- 6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
- 7.Now We create Formula Field For That Select data Type As a Formula,Click Next.

Developer Edition with Agenti...SkillWalletLightning Experience | Salesf...Recently Viewed | Buyers | Sa...Crop | SalesforceAsk Gemini

orgfarm-b89b9d25e5-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lgK000007Yik9/FieldsAndRelationships/new

Begin browsing instantly; Chrome can now launch when Windows starts. Allow

Developer Edition

Search Setup

Setup Home Object Manager

SETUP > OBJECT MANAGER

Crop

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

Step 1. Choose the field type

Step 1

NextCancel

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected

Select one of the data types below.

☐ Auto Number

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☒ Formula

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary

A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☐ Lookup Relationship

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

☐ Master-Detail Relationship

Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:

- The relationship field is required on all detail records.
- The ownership and sharing of a detail record are determined by the master record.
- When a user deletes the master record, all detail records are deleted.

31°C Mostly cloudy

Search

ENG IN

19:36:24 23-09-2026

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

The screenshot shows the Salesforce Setup interface in a web browser. The browser tabs include 'Developer Edition with Agent...', 'SkillWallet', 'Lightning Experience | Sales...', 'Recently Viewed | Buyers | Sales', 'Crop | Salesforce', and 'Ask Gemini'. The address bar shows the URL: 'orgfarm-b89b9d25e5-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lgK000007Yik9/FieldsAndRelationships/new'. A Chrome notification bar at the top says 'Begin browsing instantly: Chrome can now launch when Windows starts.' with an 'Allow' button.

The Salesforce Setup interface is displayed. The left sidebar shows the 'Setup' menu with 'Object Manager' selected. The main content area is titled 'Crop' and shows 'Step 3. Enter formula' of a 5-step process. The 'Fields & Relationships' section is active in the left sidebar. The main area contains instructions: 'Enter your formula and click Check Syntax to check for errors. Click the Advanced Formula subtab to use additional fields, operators, and functions.' An example formula is shown: 'Gross Margin = Amount - Cost__c'. Below this, there are tabs for 'Simple Formula' and 'Advanced Formula'. The 'Simple Formula' tab is selected. It shows 'Select Field Type' as 'Crop' and 'Insert Field' as 'Insert Merge Field --'. The formula entered is 'Expected Yield per Acre (Currency) = Actual_Yield__c / Total_Acres_Harvested__c'. There are buttons for 'Use Formula Assistant' and 'Insert Operator'. A 'Quick Tips' box on the right lists 'Getting Started' and 'Operators & Functions'. At the bottom, there are 'Previous', 'Next', and 'Cancel' buttons. The bottom of the screen shows a Windows taskbar with the date '23-09-2026' and time '19:39:47'.

Activity 3:

Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label(Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label(State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields&Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

Activity 3:

Creation of fields for the Farmer objects:

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3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

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5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

The screenshot displays the Salesforce Setup interface. The left sidebar shows the navigation menu with 'Setup' at the top, followed by 'Home' and 'Object Manager'. The 'Object Manager' section is expanded, showing a list of objects. The 'Farmer' object is selected, and the 'Fields & Relationships' tab is active. The main content area is titled 'New Field Dependency' and contains instructions on how to create a dependent relationship. Below the instructions, there are two steps: Step 1 (Select a controlling field and a dependent field) and Step 2 (Edit the filter rules). The 'Controlling Field' dropdown is set to 'None' and the 'Dependent Field' dropdown is also set to 'None'. A red box highlights these two dropdowns. To the right of each dropdown are 'Continue' and 'Cancel' buttons.

Developer Edition with Agent...SkillWalletLightning Experience | Salesfo...Recently Viewed | Buyers | Sai...Home | SalesforceAsk Gemini

orgfarm-b89b9d25e5-dev-ed.develop.my.salesforce-setup.com/lightning/setup/null/page?address=%2Fp%2Fdependency%2FeditDependencyUI%2Fe%3FretURL%3D%252Fsetup%252FObject...

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

Developer Edition

Search Setup

SetupHomeObject Manager

Quick Find

Setup HomeSalesforce GoService Cloud ReportsCommerce Setup AssistantHyperforce AssistantRelease UpdatesSalesforce Mobile AppLightning UsageAgentExchange OffersSales Cloud EverywhereADMINISTRATIONUsersDataEmail

SETUP

To change multiple cells at once, select multiple cells and then click the Include Values or Exclude Values button to change the visibility of all selected cells at once.

Excluded Value

Included Value

Click button to include or exclude selected values from the dependent picklist:

Include Values

Exclude Values

Showing Columns: 1 - 3 (of 3) < Previous | Next > | View All | Go to

State:	Telangana	Andhra Pradesh	Tamilnadu
District:	Vellore	Vellore	Vellore
	Thiruvannamalai	Thiruvannamalai	Thiruvannamalai
	Polur	Polur	Polur
	Ranipet	Ranipet	Ranipet
	Arcot	Arcot	Arcot

Showing Columns: 1 - 3 (of 3) < Previous | Next > | View All

Click button to include or exclude selected values from the dependent picklist:

Include Values

Exclude Values

Save

Cancel

Preview

PLATFORM TOOLS

31°C Mostly cloudy

Search

ENG IN

19:55:10 23-09-2026

7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

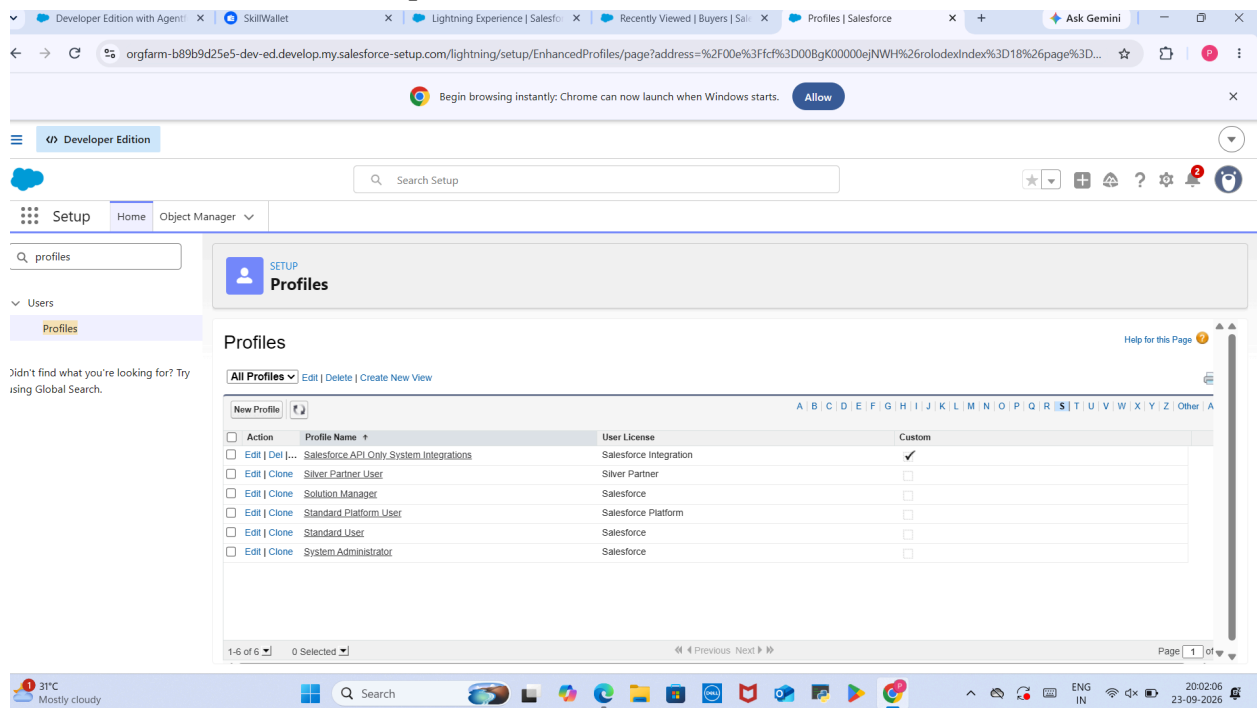
A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.
A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

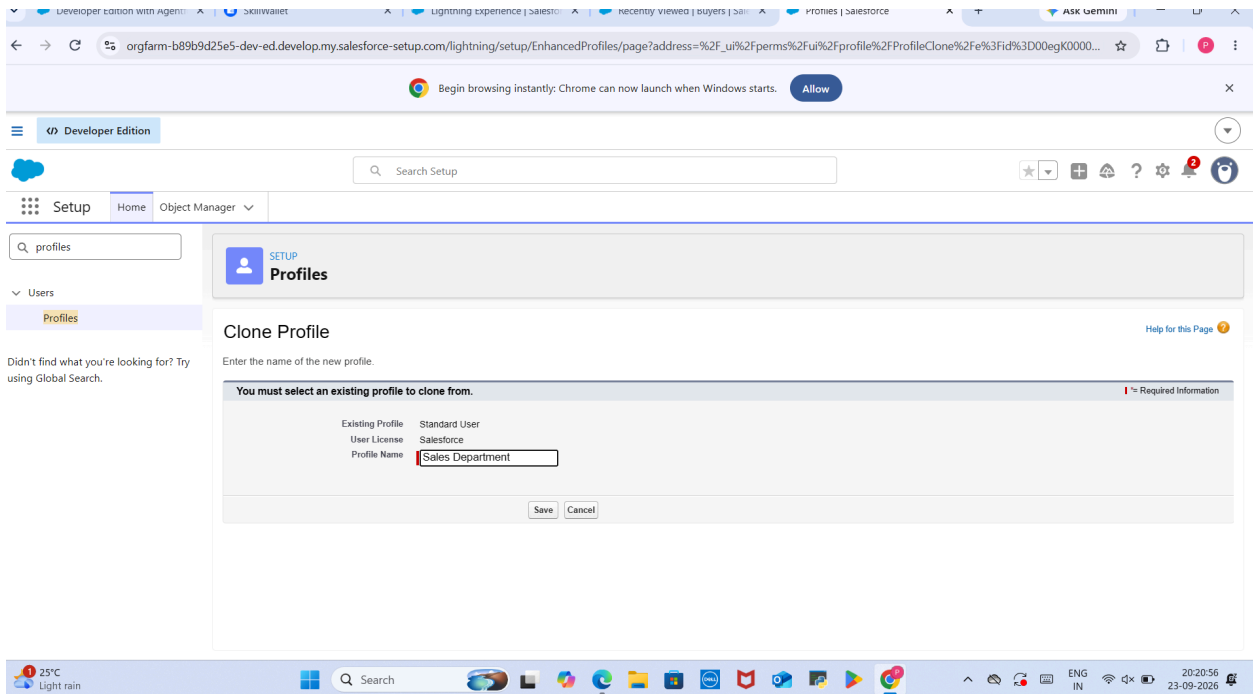
Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.



The screenshot shows the Salesforce Setup interface. The 'Profiles' page is displayed, showing a list of profiles. The 'Standard User' profile is selected, and the 'Clone' button is visible. The 'Sales Department' is entered in the 'Profile Name' field.

Action	Profile Name	User License	Custom
☐ Edit Delete	Salesforce API Only System Integrations	Salesforce Integration	☒
☐ Edit Clone	Silver Partner User	Silver Partner	☐
☐ Edit Clone	Solution Manager	Salesforce	☐
☐ Edit Clone	Standard Platform User	Salesforce Platform	☐
☐ Edit Clone	Standard User	Salesforce	☐
☐ Edit Clone	System Administrator	Salesforce	☐



5. Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

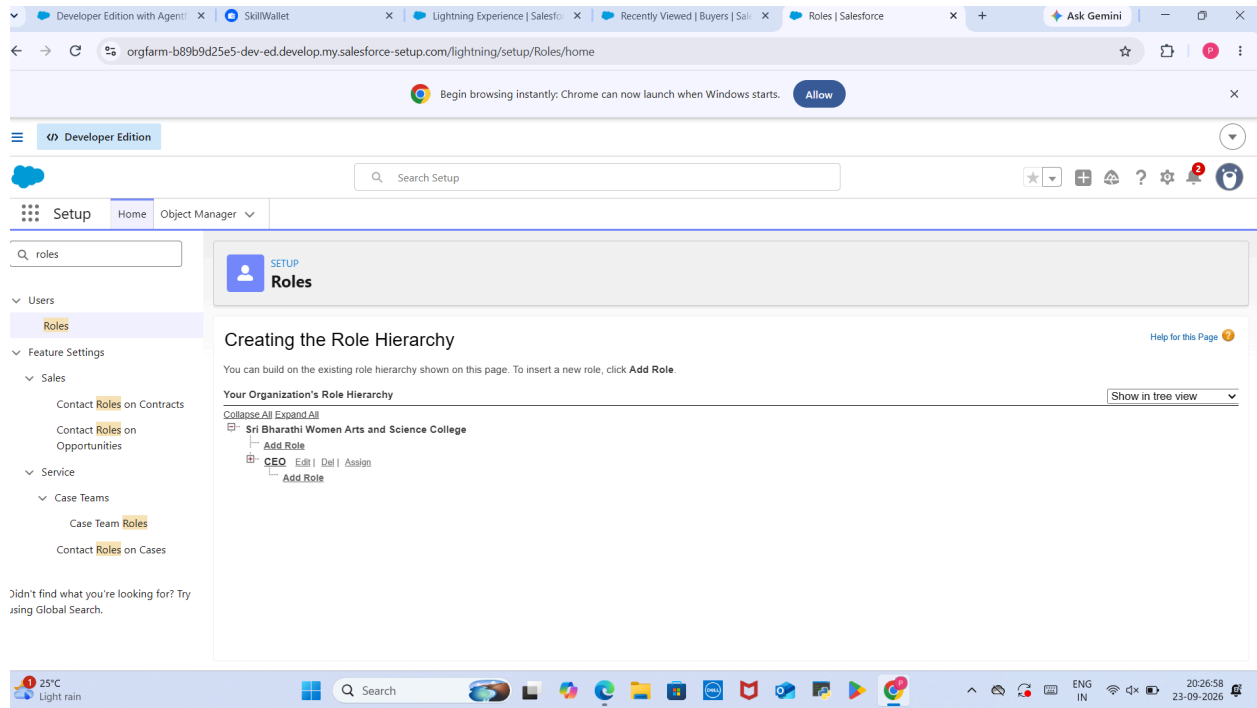
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option

The screenshot displays the Salesforce Setup interface for the 'Roles' section. The left sidebar shows the 'Setup' menu with 'Roles' highlighted under 'Users'. The main content area is titled 'SETUP Roles' and includes a search bar for 'roles'. Below the search bar, there's a section for 'Sample Role Hierarchy' with a dropdown menu set to 'Territory-based Sample'. The hierarchy diagram shows a top-level 'Executive Staff' role (CEO, President, CFO, VP, Sales) which branches into three regional director roles: 'Western Sales Director', 'Eastern Sales Director', and 'International Sales Director'. Each director role further branches into regional sales representative roles: 'Western Sales Rep' (CA, OR), 'Eastern Sales Rep' (NY, MA), and 'International Sales Rep' (Asian, European). To the right of the hierarchy diagram, there are three columns of permissions and restrictions for each role level, such as 'View & edit data, roll up forecasts, & generate reports for all users below' for the Executive Staff role. At the bottom right of the hierarchy diagram, there is a 'Set Up Roles' button and a checkbox for 'Don't show this page again'.

3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

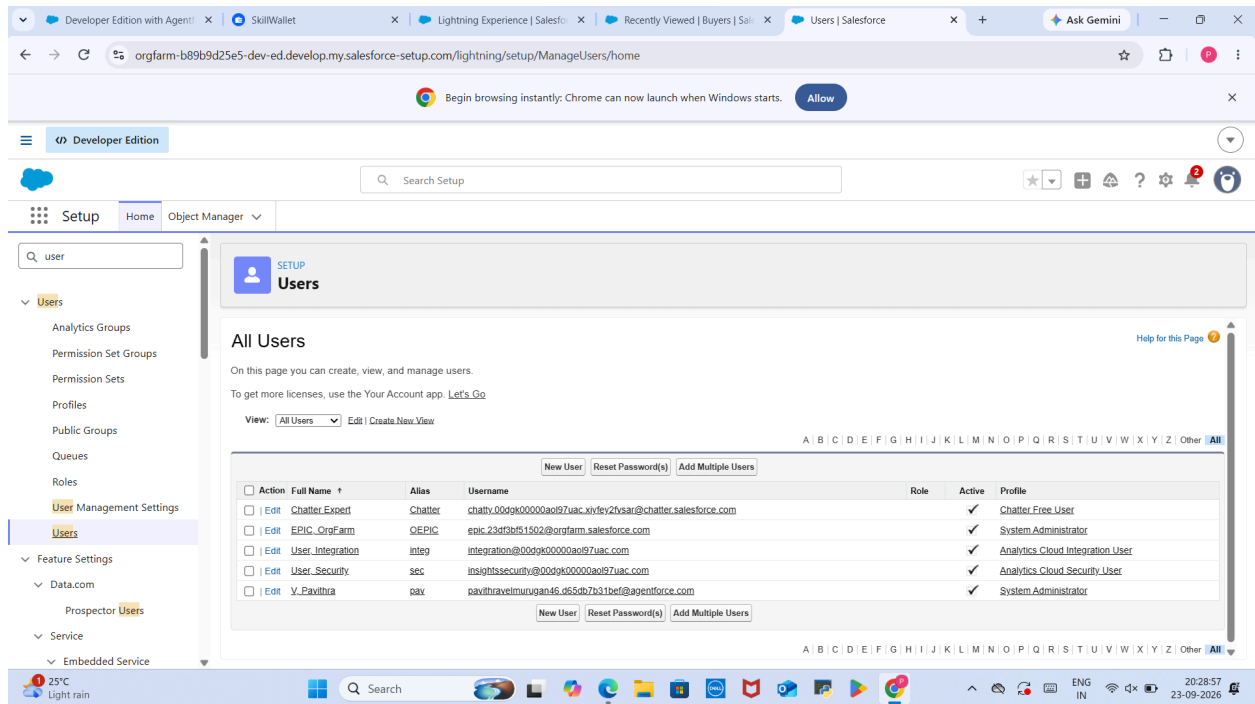
Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

The screenshot shows the Salesforce 'Sharing Settings' page in a web browser. The page is titled 'Sharing Settings' and is part of the 'Setup' menu. The left sidebar shows the 'Security' section with 'Sharing Settings' selected. The main content area is divided into three steps:

- Step 1: Select your rule type**
 - Label:
 - Rule Name:
 - Description:
- Step 2: Select your rule type**
 - Rule Type: ☐ Based on record owner ☒ Based on criteria
- Step 3: Select which records to be shared**
 - Criteria: A table with columns 'Field', 'Operator', and 'Value'.

Field	Operator	Value
Can Join Immediately	equals	true
--None--	--None--	
--None--	--None--	
--None--	--None--	
--None--	--None--	
 - Additional Options: ☒ Include records owned by users who can't have an assigned role

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows a web application interface for 'Developer Edition' with a sidebar menu. The main content area displays a modal form titled 'Information' for creating a new Farmer record. The form has two main sections: 'Farmer Name' and 'Team'. The 'Farmer Name' section includes fields for 'Farmer Name' (Kavin), 'Contact Number' (9600472552), 'State' (Tamilnadu), 'District' (Vellore), and 'Village' (Kaniyambadi). The 'Team' section shows a list of team members: Priyatharshini, Monisha, Nabitha, Pavithra, and Subiksha. The form has a 'Cancel' button, a 'Save & New' button, and a 'Save' button. The background shows a sidebar with 'Farm Management System' and 'Recently Viewed' items.

Milestone 11-Reports:

What are Reports?

A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

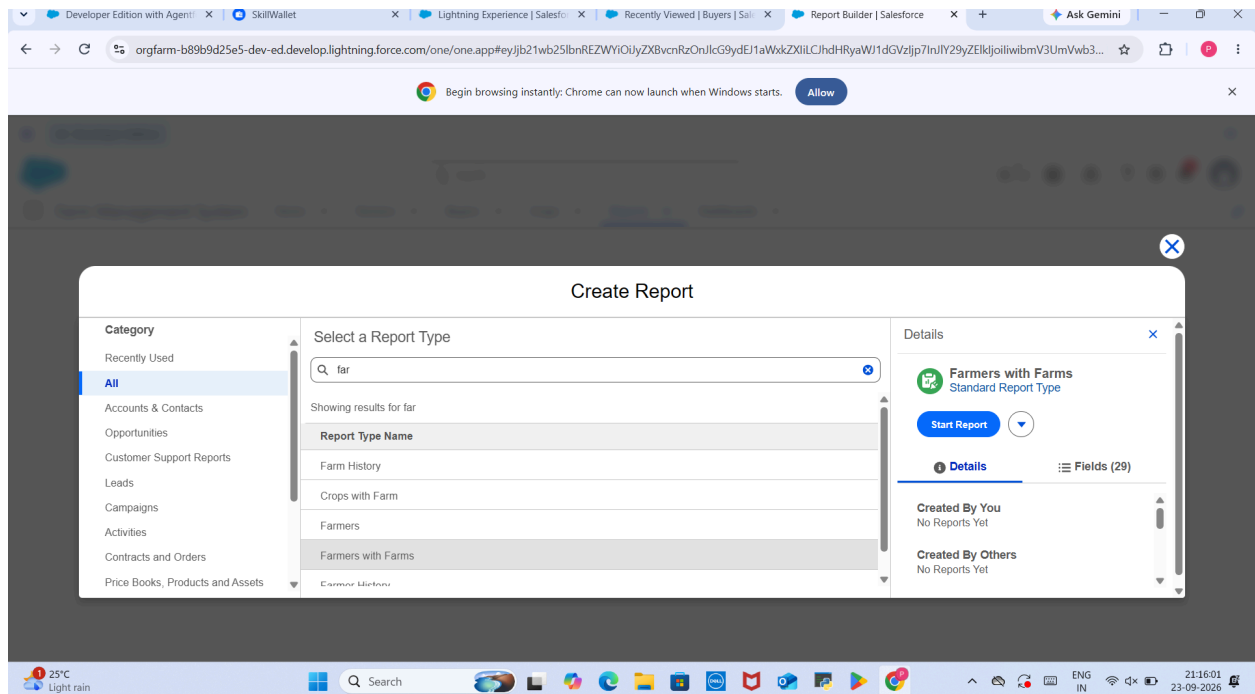
Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

1) Click on app launcher search for reports.

2) Click on the new report and select the category has Farmer with Farms.

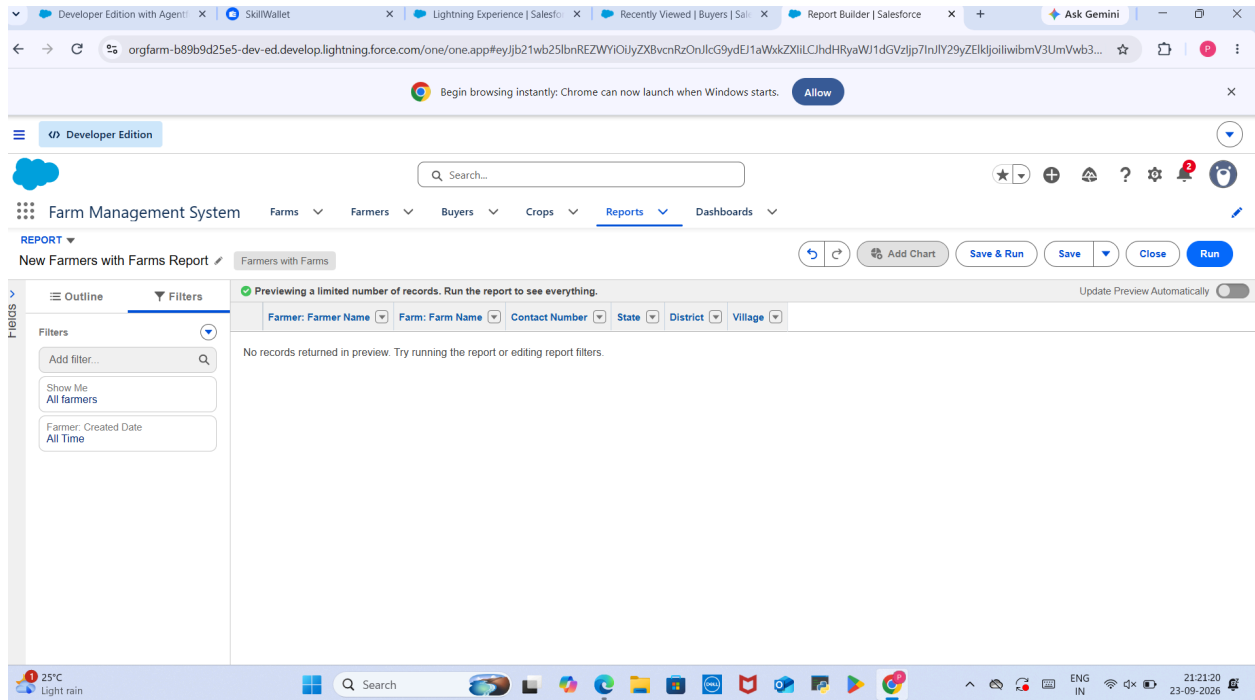


3) In the details section select the option start report.

4) In Outline Click on Column Select Which Field you want to display in the Report.

5) Click Group, Select Which Fields You Want to Group.

6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

Milestone 12-Dashboard:

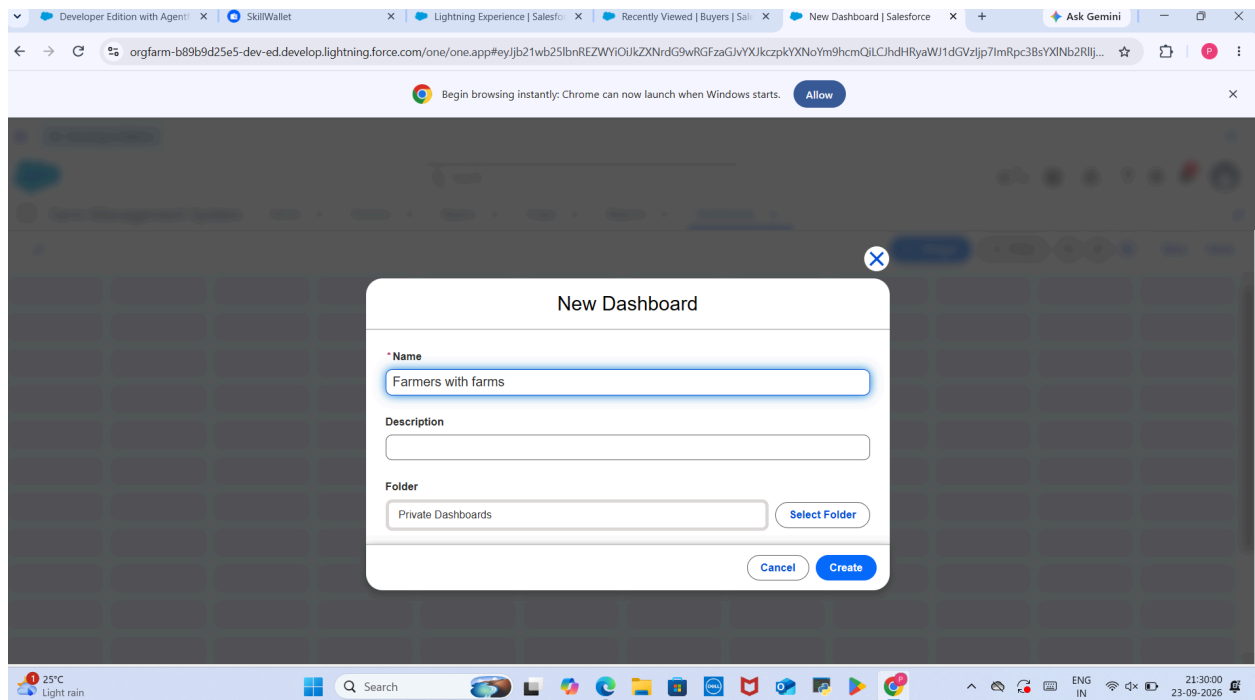
Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.

7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



FARM MANAGER

Farms

Crops

Farmers

Buyers

Reports

Dashboards

Search...



Report: Farmers with Farms

Farmers with Farms Report

Enable Field Editing



Add Chart



Print



Total Records
3

Total Farm Size(Acres)
25

☐ Irrigation Type	Farmers: Farmer Name	Farm: Farm Name	Contact Number	Farm Size(Acres)	Farm Type
☐ Localized Irrigation (3)	Rajesh	Paddy Farm	999436770	10	Livestock
	Rajesh	Sugar Farm	999436770	10	Crop
	Kavita	Wheat Farm	785262778	15	Livestock
Subtotal				25	
Total (3)				25	

Row Counts



Detail Rows



Subtotal

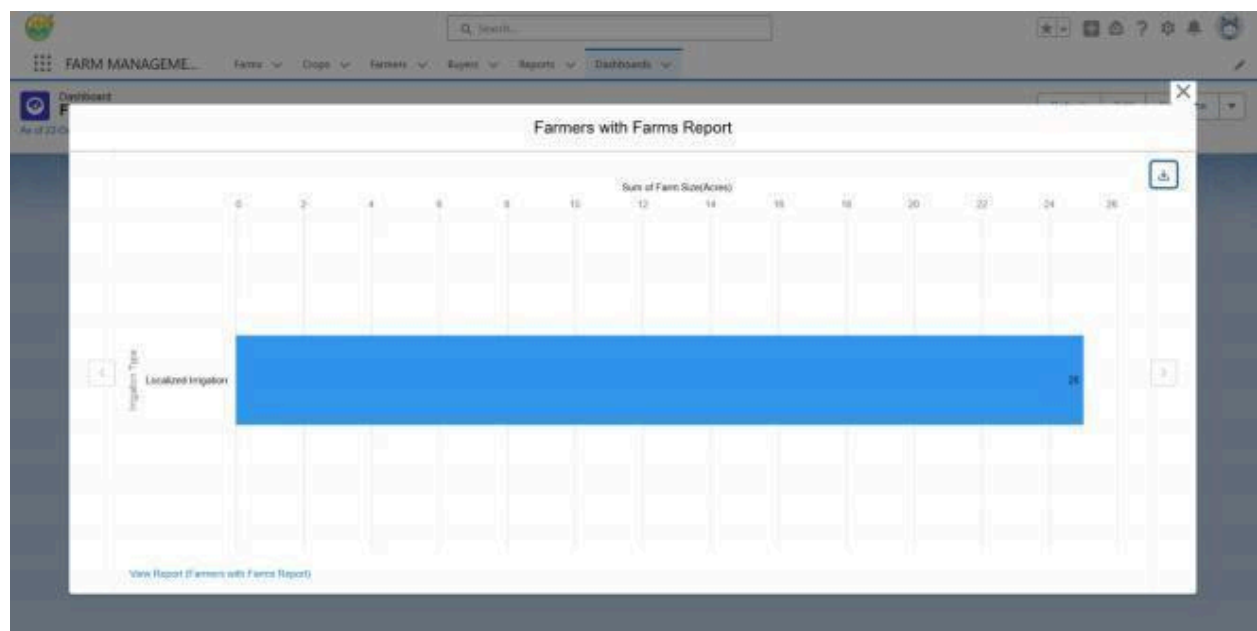


Grand Total



Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You